

Is There a Real Clinical Need?

Evidence Review: Availability of the Standardised Motor Examination in Routine Parkinson's Care

Supporting document for the TD/PIGD longitudinal confidence-modelling project | 10 Sep 2026

1. The question this document answers

Our TD/PIGD label is computed from a published formula that requires item-level MDS-UPDRS Part II and Part III scores. In PPMI, those items are collected at essentially every visit — so within PPMI, the label is directly computable and there is nothing to "predict." The project therefore only has a clinical justification if the standardised motor examination is *not* routinely available outside research cohorts. This document tests that assumption against published evidence.

2. Finding 1 — The MDS-UPDRS is a research instrument, not a routine-care one

Dimension	Published finding
Length / burden	50 items; ~30 minutes to administer in full
Part III specifically	33 scored items from 18 questions, with left/right/body-region breakdowns
Rater requirement	MDS runs a formal MDS-UPDRS Certificate Program; the Part III teaching programme alone runs ~2 hours
Actual frequency of use	"Not always administered at every clinical visit"; typically every 6 or 12 months, and mainly within large longitudinal research cohorts
Telehealth	Even fewer MDS-UPDRS ratings are performed remotely — hard to visualise body parts and performance tasks

The scale's own literature repeatedly names administration time and complexity as the reason it is not adopted in ordinary clinics — this is the stated motivation behind published efforts to build abbreviated and remote versions of the scale.

3. Finding 2 — Specialist access in India (n=267 clinicians, 23 states/UTs)

A 2025 national cross-sectional survey published in the *Journal of Movement Disorders* gives the clearest picture of who actually delivers PD care in India:

Metric	Value
Initial diagnosis made by a movement disorder specialist	12.6% of practices
Follow-up care by a movement disorder specialist	19.5% of practices
Follow-up care by a neurologist	85.7%

Follow-up care by internist-equivalent doctors	48.9%
Follow-up care by MBBS (general) doctors	7.4%
Movement disorder specialist "rarely accessible locally"	79.2% of respondents
PD nurse "rarely accessible locally"	94.4% of respondents
Respondents practising in urban areas	92.1% (urban:rural doctor density gap 3.8:1)
Patients paying fully out-of-pocket (>75% of caseload)	55.4% of practices

Context: India holds roughly 9.5% of the global PD population (~576,000 people as of the GBD 2016 estimate). A 2015 analysis of professional-society membership found that **no neurologist or neurosurgeon resided in regions covering 934.8 million people**, with only 3% of neurologists serving rural areas home to 84.6 million people.

Implication: in the large majority of Indian PD encounters, care is delivered by clinicians who are not movement-disorder specialists and are unlikely to be MDS-UPDRS certified. A subtype label that formally requires certified item-level Part III rating is, in practice, unavailable to most of these patients.

4. Finding 3 — Even in the UK, review ≠ standardised motor exam

NICE guideline NG71 recommends that a Parkinson's diagnosis be reviewed regularly — every 6 to 12 months — and that suspected cases be referred untreated to a specialist. Note what this does and does not say: it specifies review *frequency*, not that a full item-level MDS-UPDRS is scored at each review. The UK Parkinson's Audit, which measures compliance against these standards, documents substantial variability in access to the wider multidisciplinary team (e.g. 31% of people were unable to access a physiotherapist, or unsure whether they could, pre-pandemic).

So even in a well-resourced national health system with an explicit guideline, the operative standard is periodic specialist review — not a certified, item-level motor examination captured in structured form at each contact. The research-grade data our formula needs is not a by-product of ordinary UK care either.

5. Finding 4 — The complication that reshapes our framing

The same Indian survey undercuts the naive version of our argument. If the pitch were "replace the motor exam with imaging and genetics," the data says those inputs are **even less** available than the exam:

Investigation	Reported practice in India
DaTscan / dopamine transporter SPECT	94.4% of clinicians rarely advise it
Genetic testing	94.9% rarely advise it
CT scan	75.7% rarely advise it
MRI	71.5% frequently recommend it

A physical examination needs a trained clinician and no equipment. A DaTscan needs a gamma camera and a radiotracer; genotyping needs a lab. In low-resource settings the exam is the *cheap* input and imaging/genetics are the expensive ones — so any framing of the form "predict the subtype from imaging and genetics so you don't need the exam" is backwards, and a reviewer from the field would say so immediately.

6. The framing that survives the evidence

There is a sharper distinction hiding inside the MDS-UPDRS itself, and it is the one worth building on:

Component	Who supplies it	Access requirement	Availability in our PPMI data
Part II (Motor Aspects of Daily Living)	Patient-reported questionnaire	No specialist required; can be self-completed or collected by any staff member, on paper or remotely	NP2TRMR, NP2WALK, NP2FREZ — 99.9–100% complete in our extract
Part III (Motor Examination)	Clinician-administered exam	Requires a trained/certified rater, in person, ~30 min for the full scale	13 tremor + gait items — 94.1–94.2% complete in our extract

Crucially, Part II contains its own tremor item **and** its own walking/freezing items — the same two clinical dimensions the TD/PIGD ratio is built from — but sourced from the patient rather than from a certified examiner. This makes a concrete, falsifiable research question available to us with the data we already hold:

Can the TD/PIGD/Indeterminate classification — defined by the full Part II + Part III Stebbins formula — be recovered from patient-reported and low-cost inputs alone, without the specialist-administered Part III exam? And with what confidence, per patient?

This version of the project: (a) uses the formula-derived label as trustworthy ground truth, exactly as the renewed approach intends; (b) makes the prediction task genuinely non-trivial, because the model is denied the very items that define the label; (c) has a real-world beneficiary supported by the evidence above — the ~80% of Indian PD encounters without movement-disorder-specialist involvement, and any setting where a certified rater or a 30-minute exam slot is not available; and (d) leaves the conformal confidence and transition-risk machinery completely intact, since "how sure are we without the exam" is precisely a confidence question.

7. Honest limitations of this argument

- The India survey reports what clinicians say they do, not audited chart data — self-report bias applies, and 92.1% of respondents were urban, so rural practice is under-represented rather than directly measured.
- None of the sources found state a direct measured figure for "% of routine PD visits worldwide at which a complete item-level MDS-UPDRS is scored." Our claim is assembled from adjacent evidence (administration burden, certification requirement, specialist scarcity, guideline wording), not from one clean statistic. That gap should be acknowledged rather than papered over.
- PPMI participants are early-stage, largely from specialised centres in higher-income settings, and are not representative of Indian or rural patients — so a model trained here would need external validation before any claim of usefulness in those settings.
- Part II is patient-reported and therefore carries its own biases (recall, self-assessment, literacy, language) that a certified examiner's rating does not — 'cheaper to collect' is not the same as 'equally reliable'.

8. Sources

- Rajan R, Kumar H, Radhakrishnan DM, et al. "The Landscape of Parkinson's Disease Treatment in India: A National Cross-Sectional Survey of Clinical Practitioners." *J Mov Disord.* 2025;19(2):212–216. doi:10.14802/jmd.25201 — <https://pmc.ncbi.nlm.nih.gov/articles/PMC13175737/>
- NICE Guideline NG71, "Parkinson's disease in adults" — <https://www.nice.org.uk/guidance/ng71/chapter/recommendations>
- UK Parkinson's Audit, Parkinson's UK — <https://www.parkinsons.org.uk/professionals/audit>
- Parkinson's UK written evidence to Parliament (access/waiting times) — <https://committees.parliament.uk/writtenevidence/113679/pdf/>
- "Abbreviated MDS-UPDRS for Remote Monitoring in PD Identified Using Exhaustive Computational Search" (administration burden, telehealth limits) — <https://pmc.ncbi.nlm.nih.gov/articles/PMC9197633/>
- MDS-UPDRS Training / Certificate Program, International Parkinson and Movement Disorder Society — <https://mds.movementdisorders.org/updrs/>
- Ganapathy K. "Distribution of neurologists and neurosurgeons in India and its relevance to the adoption of telemedicine." *Neurol India.* 2015;63:142–154 (cited within the JMD survey above).
- Item-level completeness figures are computed directly from our own PPMI extract (MDS-UPDRS Part II n=37,692 rows; Part III n=39,255 rows).